

PROJECT OVERVIEW

Nonlinear Multifactor Scoring

AI-Assisted Rules-to-Code Development

An independent public demonstration of turning a decision framework into a transparent, repeatable candidate-priority ranking.

DETERMINISTIC JAVASCRIPT

SIX INPUTS + ONE GATE

LOCAL BROWSER EXECUTION

What this project demonstrates

The score combines diminishing benefit, readiness, cost, benefit-readiness interaction, time decay, risk attenuation and data quality. Eligibility is checked separately so an ineligible candidate cannot enter the ranking.

MY ROLE

Define needs, scoring rules and acceptance criteria; steer AI-generated implementation and revisions; review behavior against the intended rules.

AI COLLABORATION

AI assists with producing and revising code, tests and documentation. The resulting scoring calculation is fixed JavaScript with no runtime AI calls.

Six numerical inputs

01 Benefit

Expected value in the public rule set

02 Readiness

Feasibility or execution readiness

03 Cost

Higher values reduce preference

04 Risk

Higher values attenuate the score

05 Data quality

Quality multiplier and eligibility floor

06 Age

Elapsed days; older inputs decay

Public-demo boundary: the rules, parameters and examples were constructed separately for this demonstration. Private original rules, model details and data are excluded.

CALCULATION SPECIFICATION

From rules to a traceable score

One explicit calculation, with validation and eligibility ahead of scoring.

1. Validate policy and inputs

Benefit, readiness, cost, risk and quality are on a 0-100 scale; divide each by 100 to obtain **u, m, c, r, q**. Age **t** is 0-365 days. Validate before applying the prerequisite gate.

The demo UI accepts $0.1 \leq k \leq 10$ and half-life $0.25 \leq H \leq 365$ days. Weights, interaction coefficient **gamma** and risk coefficient **beta** are non-negative; total normalization weight **Z** > 0.

2. Enforce eligibility

Prerequisite = true AND raw quality ≥ 60 AND raw risk ≤ 80 .

Otherwise return a null score and exclude the candidate from rankings.

3. Transform benefit, time and risk

BENEFIT SATURATION

$$U = (1 - \exp(-k \cdot u)) / (1 - \exp(-k))$$

Normalizes benefit to 0-1 while reducing marginal gains.

TIME DECAY

$$D = \exp(-\ln(2) \cdot t / H)$$

A half-life H halves the time multiplier every H days.

RISK ATTENUATION

$$P = \exp(-\beta \cdot r^2)$$

Applies a nonlinear penalty as normalized risk rises.

4. Combine and normalize

$$Z = w_u + w_m + w_c + \gamma$$

$$B = (w_u \cdot U + w_m \cdot m + w_c \cdot (1 - c) + \gamma \cdot U \cdot m) / Z$$

$$S = 100 * B * D * P * q$$

Default policy: $k = 3$; $H = 14$ days; $\beta = 2$; $\gamma = 0.2$;
 $w_u = 0.5$, $w_m = 0.3$, $w_c = 0.2$; therefore $Z = 1.2$.

The score is a policy-defined priority index, not a calibrated probability.

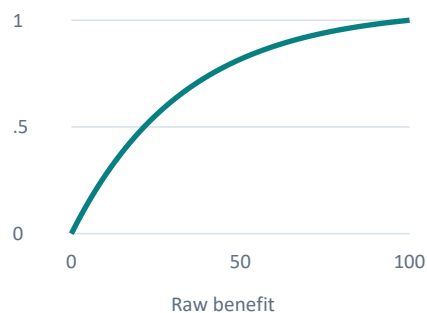
BEHAVIOR AND CALCULATION TRACE

See the rules change the ranking

Illustrative cases under a fixed public policy; all prerequisite flags are true.

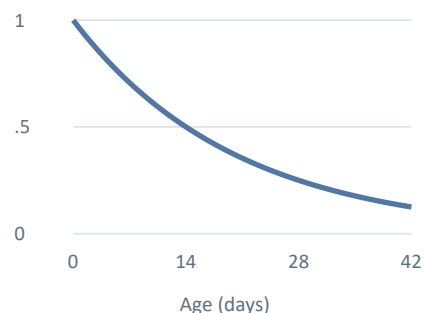
Benefit saturation

$k = 3$; U



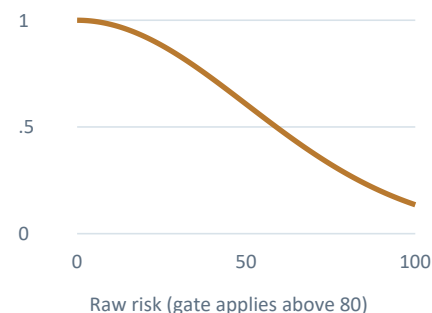
Time decay

$H = 14$ days; D



Risk attenuation

$\beta = 2$; P



Default-policy results

Case	Benefit	Ready	Cost	Risk	Quality	Age (d)	Score
A	95	85	30	65	90	21	12.07
B	90	90	30	15	97	7	58.88
C	65	70	15	20	92	1	64.51
D	98	95	20	10	45	1	Excluded

Ranking: $C > B > A$. Case D is excluded because quality 45 is below 60.

A parameter change can reverse priorities

Half-life	B score	C score	Higher-ranked
7 days	41.63	61.39	C
14 days	58.88	64.51	C
30 days	70.83	66.23	B

CASE C / WORKED TRACE

$U = 0.902667$ $B = 0.798089$ $D = 0.951695$

$P = 0.923116$ $q = 0.92$ $Z = 1.2$

$S = 100 \times B \times D \times P \times q = 64.505019 \rightarrow 64.51$

Arithmetic contributions: benefit 30.3989 + readiness 14.1443 + cost 11.4501 + interaction 8.5117. This decomposition is not causal attribution.

VERIFICATION, LIMITS AND WORKING PREFERENCES

Evidence with clear boundaries

A portfolio demonstration of rule implementation and review, with claims kept within the available evidence.

48 / 48**In-page checks rerun successfully.**

Selected outputs were also independently recomputed from the displayed formulas.

What the review supports

- All 48 in-page checks were rerun and passed. Eight targeted UI probes checked invalid parameters, quality/risk eligibility boundaries, gamma and half-life changes.
- Default example rankings, a selected calculation trace, half-life sensitivity and the normalized interaction effect were independently recalculated.
- Review covered the displayed evaluation function, visible test results and selected behavior. It was not a full source-code security audit or a real-world outcome assessment.

Interpretation and limitations

- No model training, calibrated success probability, demonstrated predictive accuracy or production deployment is claimed.
- A larger interaction coefficient does not guarantee a larger score: gamma also increases Z. For Case C, gamma = 0 gives 67.19; gamma = 0.2 gives 64.51.
- Parameter sensitivity illustrates policy behavior. It does not establish model improvement or evidence of better real-world decisions.

CURRENT WORKING PREFERENCES**Fully remote, part-time / project-based**

Weekdays 19:00-22:00 Taiwan time (UTC+8).

I am not currently accepting engagements that require meetings conducted entirely in English.